1.A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Answer

The lag observed while switching between applications is mainly caused by delays in frequently required instructions and data. The processor uses a memory hierarchy consisting of L1, L2, L3 cache and DRAM to reduce the average memory access time.

Role of Different Memories

Memory	**Role**	**Speed**	**Capacity**
L1 Cache	Stores the most frequently used instructions and data	Very High	Very Small
L2 Cache	Stores data not available in L1	High	Small
L3 Cache	Shared cache for multiple CPU cores	Moderate to High	Larger
DRAM	Stores running applications and large data	Lower than cache	Large

L1 cache is the fastest memory and is located closest to the processor. If the required data is not available in L1, the processor checks L2 and then L3 cache. If the data is absent from all caches, the processor accesses DRAM. Since DRAM has considerably higher latency than cache, frequent DRAM accesses can cause delays during application switching.

Proposed Hierarchical Memory Redesign

The proposed hierarchy is:

CPU → L1 Cache → L2 Cache → L3 Cache → High-Speed DRAM

The manufacturer should increase the efficiency and capacity of L2 and L3 caches and use high-bandwidth LPDDR5X or newer mobile DRAM. Cache prefetching can also be used to predict and load frequently required application data before it is requested.

Expected Improvement

The redesigned hierarchy increases cache hit rate and reduces the number of accesses to DRAM. Therefore, memory access latency decreases and data throughput increases. This results in faster application switching, smoother multitasking and improved smartphone performance.

Performance Analysis

The average memory access time can be represented as:

AMAT = Hit Time + Miss Rate × Miss Penalty

By increasing cache hit rates, the number of accesses reaching DRAM is reduced. Therefore, application switching becomes faster and CPU throughput increases.

Tools and Technologies

- CPU/cache simulator
- ARM-based processor architecture
- Cache profiling tools
- Performance monitoring counters
- LPDDR memory technology

Conclusion

A balanced hierarchy using fast L1 and L2 caches, a sufficiently large shared L3 cache and high-bandwidth DRAM provides low latency and high throughput. This effectively reduces lag during application switching.

2. A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Answer

High database-query latency can occur when frequently accessed database data is not available in the CPU cache or main memory. The processor may then need to access DRAM or, in the case of a page fault, secondary storage.

Comparison of Cache Hierarchy and Virtual Memory Paging

Feature	**Cache Hierarchy**	**Virtual Memory Paging**
Purpose	Reduces CPU memory-access latency	Provides larger logical memory space
Main Technology	SRAM	DRAM and SSD
Access Speed	Extremely Fast	Slower
Data Unit	Cache Line	Page
Controlled By	Hardware/cache controller	Operating System and MMU
Main Problem	Cache Miss	Page Fault
Main Benefit	Faster CPU execution	Supports large applications

The cache hierarchy consists of L1, L2 and L3 caches. Frequently accessed database indexes and instructions can benefit from high cache hit rates.

Virtual memory uses DRAM as physical memory and secondary storage such as SSD when required data is not currently available in RAM. A page fault causes data to be transferred from storage to DRAM, resulting in significantly higher latency.

Recommended Memory Architecture

CPU → L1 → L2 → L3 → DDR5 DRAM → NVMe SSD

The cloud provider should:

1. Increase DRAM capacity to reduce page faults.
2. Use high-speed DDR5 DRAM.
3. Optimize database buffer and cache management.
4. Increase CPU cache hit rates through efficient data organization.
5. Use high-speed NVMe SSDs for persistent storage and paging.
6. Reduce unnecessary swapping and paging.
7. Keep frequently accessed database indexes and data in DRAM.

Expected Improvement

Increasing DRAM capacity and optimizing cache usage reduces cache misses, page faults and storage accesses. This decreases database-query latency and improves overall server throughput.

Performance Analysis

Database latency can be reduced significantly by minimizing:

- L3 cache misses
- DRAM accesses
- Page faults
- Disk I/O operations

The most important improvement is to provide enough DRAM so that the active database working set remains in physical memory.

Tools

- Linux vmstat
- perf
- Database performance-monitoring tools
- NUMA monitoring tools
- NVMe monitoring utilities

Conclusion

Cache hierarchy is essential for reducing CPU access latency, while virtual memory is useful for extending available memory capacity. For cloud database servers, the best solution is to maximize cache efficiency and DRAM capacity while using NVMe SSDs as secondary storage.

3. An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Answer

An autonomous vehicle continuously receives large amounts of data from cameras, LiDAR, radar, GPS and other sensors. This data must be processed with very low latency to support real-time decisions. Therefore, the memory system must provide both high bandwidth and fast access.

Comparison of SRAM, DRAM and MRAM

Feature	SRAM	DRAM	MRAM
Speed	Very High	High	High
Volatility	Volatile	Volatile	Non-Volatile
Capacity	Small	Large	Medium
Latency	Very Low	Higher than SRAM	Low
Power	Relatively High	Moderate	Low standby power
Main Application	Cache	Main Memory	Persistent/Critical Data

Evaluation

SRAM:
SRAM provides extremely fast access and very low latency. It is suitable for CPU and AI accelerator caches. However, it has high cost and limited capacity.

DRAM:
DRAM provides large capacity and high bandwidth. It is suitable for storing large sensor datasets, image frames and intermediate AI-processing data.

MRAM:
MRAM is non-volatile and provides fast access with low standby power. It is suitable for storing critical configuration information, system parameters and data that must survive power interruptions.

Proposed Memory Hierarchy

Sensors → SRAM Cache → High-Bandwidth DRAM → MRAM → Flash/SSD

SRAM should store the most frequently accessed real-time data. High-bandwidth DRAM should store large sensor datasets and AI inference data. MRAM should store persistent and critical information.

Expected Improvement

The use of SRAM reduces processing latency, while high-bandwidth DRAM supports large-scale sensor-data processing. MRAM provides non-volatile storage with low standby power consumption.

Performance Analysis

Autonomous driving requires both low latency and high bandwidth. SRAM provides the lowest latency, while DRAM provides much greater capacity. Therefore, using only SRAM would be expensive and impractical, while using only DRAM would increase latency.

The proposed hybrid hierarchy provides:

Low latency → SRAM

High bandwidth and capacity → DRAM

Non-volatile fast storage → MRAM

This balance supports real-time sensor processing.

Tools

- Hardware performance counters
- Cache and memory simulators
- Sensor-data profiling
- DMA performance monitoring
- AI accelerator profiling tools

Conclusion

A hierarchical system combining SRAM, high-bandwidth DRAM and MRAM provides a good balance between latency, bandwidth, capacity, power consumption and reliability. This architecture is suitable for real-time autonomous vehicle processing.

4. An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

Answer

An edge AI device needs to load machine-learning models quickly and execute inference with minimum power consumption. The memory hierarchy should combine high-capacity storage with fast and energy-efficient memories.

Comparison of NAND Flash, DRAM and RRAM

Feature	**NAND Flash**	**DRAM**	**RRAM**
Volatility	Non-Volatile	Volatile	Non-Volatile
Speed	Moderate	High	High
Capacity	Very High	High	Medium
Power Consumption	Low	Moderate	Low

Feature	NAND Flash	DRAM	RRAM
Main Use	Permanent Model Storage	Active Model Execution	Fast Non-Volatile Storage

Evaluation

NAND Flash:
NAND Flash provides high storage capacity and retains data without power. It is suitable for storing large AI models, applications and datasets. However, it has higher access latency compared with DRAM.

DRAM:
DRAM provides fast access and is suitable for storing the active AI model and intermediate inference data. However, it consumes more power and is volatile.

RRAM:
RRAM is a non-volatile memory technology that can provide fast access with potentially lower power consumption. It can be used for frequently accessed model parameters and fast model storage.

Proposed Memory Hierarchy

AI Processor → On-Chip SRAM → DRAM → RRAM → NAND Flash

The most frequently accessed data should be stored in fast on-chip memory. DRAM should hold the active model during inference. RRAM can store frequently used model parameters in a non-volatile layer, while NAND Flash should store large models and less frequently used data.

Expected Improvement

The proposed hierarchy reduces model-loading time and unnecessary accesses to NAND Flash. It also reduces power consumption by using fast memories for frequently accessed data.

Performance Analysis

The main objective is to minimize the time required to move model data from storage to the AI accelerator.

Instead of:

NAND Flash → CPU/AI Accelerator

the proposed system uses:

NAND Flash → RRAM/DRAM → SRAM/AI Accelerator

This reduces repeated accesses to slower NAND storage and improves inference performance.

Tools

- AI accelerator profiler
- Memory bandwidth profiler
- Power monitoring tools
- Model quantization tools

- Hardware simulation tools

Conclusion

A combination of fast on-chip memory, DRAM, RRAM and high-capacity NAND Flash provides a balance between speed, capacity and power efficiency. This architecture is suitable for low-power edge AI inference devices.

5. A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Answer

Frame drops during large scene loading occur when the CPU and GPU cannot receive game data, textures, models and other assets quickly enough. The performance depends strongly on cache latency and DRAM bandwidth.

Comparison of L3 Cache and DRAM

Feature	**L3 Cache**	**DRAM**
Technology	SRAM	DRAM
Latency	Very Low	Higher
Capacity	Small	Large
Bandwidth	Very High for cache access	High
Cost per Bit	High	Lower
Main Use	Frequently accessed data	Large game assets

L3 cache is a large, fast cache shared among processor cores. It stores frequently accessed instructions and data and reduces the number of accesses to DRAM.

DRAM has much greater capacity than L3 cache and is therefore used to store large game assets such as textures, character models, game environments and other active data. However, DRAM has higher latency than cache.

Proposed Memory Hierarchy

CPU/GPU → L1 Cache → L2 Cache → Large L3 Cache → High-Bandwidth GDDR DRAM → NVMe SSD

Memory Hierarchy Enhancements

1. Increase the effective L3 cache capacity for frequently accessed game data.
2. Use high-bandwidth GDDR memory for graphics and large scene data.
3. Implement hardware and software prefetching.

4. Stream game assets from NVMe SSD to DRAM before they are required.
5. Use data compression to reduce memory-transfer requirements.
6. Optimize game-engine memory allocation to improve cache locality.

Expected Improvement

A larger and more efficient L3 cache reduces CPU memory latency, while high-bandwidth DRAM allows large amounts of graphical data to be transferred efficiently. Prefetching and asset streaming ensure that required data is available before rendering.

Performance Analysis

Frame rate can be represented as:

FPS = 1 / Frame Time

For example, at 60 FPS:

Frame Time = 1/60 ≈ 16.67 ms

If memory stalls increase frame time beyond the available frame budget, frame drops occur.

Therefore, reducing cache misses, increasing DRAM bandwidth and improving asset streaming can help maintain a stable frame rate.

Tools

- GPU/CPU profilers
- Frame-time analyzers
- Cache-miss counters
- Memory-bandwidth monitoring
- Game-engine profiling tools
- SSD performance benchmarks

Conclusion

L3 cache provides low-latency access to frequently used data, while high-bandwidth DRAM provides the capacity required for large game scenes. Combining a larger cache hierarchy, high-bandwidth GDDR memory, efficient prefetching and NVMe-based asset streaming can reduce frame drops and improve gaming performance.

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	**CSA1223**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:	Mogana Priya	**Reg.No.:**	192321165
Department:	IT	**Date:**	

82 %

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
   A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

2. Cloud Server Memory Bottleneck
   A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

3. Autonomous Vehicle Data Processing
   An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

4. AI Inference Edge Device
   An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

5. High-Performance Gaming Console
   A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I Mogana Priya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)	
Q1	1	1	1.25	0.75	0.5	4.5
Q2	0.5	1	1.25	0.75	0.5	4
Q3	1	1.25	1	0.5	0.25	4
Q4	0.5	1.25	1	0.5	0.5	3.75
Q5	1	1	1.	0.75	0.5	4.25
						20.5

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem(1.25)	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards(1.25)	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data(0.75)	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation(0.5)	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis